

AI built it. Growth broke it. We rebuilt it.

An AI education platform serving 420,000+ educators

THE STAKES

The platform began as an AI-generated, no-code-style implementation. It worked — too well: growth outran what that architecture could ever carry, and neither the founders nor the app-builder tooling could scale it. Every week at that wall is churn and lost subscriptions. This is the story most companies shipping AI-built software are about to live: it works until it matters.

WHAT I DID

The company hired our team to rebuild it properly. Backend was **two engineers** — my team lead and me. The rebuilt platform shipped as the public 2.0 launch with 150+ AI tools:

- **Production LLM features that survive real traffic:** RAG with LangChain and vector embeddings, multi-provider routing (Azure OpenAI, AWS Bedrock), streaming endpoints
- **The revenue plumbing:** subscriptions, school management, Stripe billing, Cognito authentication
- **The scaling work:** sync-to-async migration (ASGI/Uvicorn), AWS ECS deployments, CI/CD
- **An email system admins run without engineers:** templates with live HTML preview and automated workflows, designed and built end to end

OUTCOMES

- The rebuilt platform launched publicly and serves **420,000+ educators**.
- An AI-generated prototype became a scalable production system — without the business stopping.
- I know exactly where AI-generated codebases break at scale, because I have rebuilt one that 420,000 people depend on.

WHY IT MATTERS

If your product was built fast with AI tooling and is now meeting real traffic, this is the playbook: keep the product live, rebuild the foundations underneath it, ship the launch on time.

STACK

Python · Django · LangChain · Azure OpenAI · AWS Bedrock · AWS ECS · Cognito · Stripe

This is the engineering standard behind Alpine's fixed-price delivery audit: findings with file-and-line evidence, an implementation playbook, and an execution-ready backlog your team can run with. References on request.

Stanislav Petrov · Alpine · slavpetroff@gmail.com · linkedin.com/in/stanislav-petrov-a72739105